MultiHop Optimal Time Complexity Clustering for Emerging IoT Applications

Yann Chebu[1] · **Alain B. Bomgni** *[2] · **Hafiz M. Ali** [3] · **David G. Zanfack** [4] · **Waleed Ejaz** [5] · **Clémentin Tayou D.** [6] · **Etienne G. Zohim** [7]

Received: date / Accepted: date

Abstract The Sixth-generation (6G) wireless communication networks are expected to support heterogeneous services and decentralized infrastructure with resource-aware smart self-organization for Internet of things (IoT) applications. Large-scale IoT applications face challenges like load balancing and scalability within the network due to the inherent vulnerability of ad-hoc structures. This paper proposes a multiHop constant-time complexity clustering algorithm (MultiHopFast) for IoT networks to address these challenges. The proposed MultiHopFast algorithm reduces the computing burden from IoT nodes with smart load balancing to ensure IoT network scalability. The algorithm addresses the network load, scalability, and time efficiency challenges. Using neighbourhood heuristics, the MultiHopFast algorithm builds appropriate size (i.e., up to 5 hops) of clusters with participating IoT nodes. Each cluster is associated with a cluster head (CH) (or a coordinator). The MultiHopFast algorithm probabilistically selects CH for each cluster. When compared with state-of-the-art counterparts, MultiHopFast algorithm: (i) operates with constant-time complexity in a large scale network as well as in small-scale networks, (ii) runs without any impact on network scalability, and (iii) creates 12% fewer CHs to save precious resources such as energy. Better use of heuristics and resource-aware self-organization, constant-time computational complexity, and network operation with fewer CHs demonstrate that the performance of the MultiHopFast algorithm surpasses the compared algorithms in the literature. The MultiHopFast algorithm is envisioned as a better candidate to match the standard and expectations set by the 6G wireless communications.

Keywords Clustering · IoT networks · optimization · sixth-generation · multihop clustering.

Yann B. CHEBU [1]
Department of Mathematics and Computer science, University of Dschang, Cameroon.
E-mail: yannchebu@gmail.com

Alain Bertrand Bomgni*[2] (✉)
Department of Biomedical Engineering, University of South Dakota, USA.
E-mail: alain.bomgni@usd.edu

Hafiz Munsub Ali [3]
Department of Biomedical Engineering, University of South Dakota, USA.
E-mail: hafiz.ali@usd.edu

David Gnimpieba Zanfack [4]
AGT Technology, France.
E-mail: davgnimpie@gmail.com

Waleed Ejaz [5]
Department of Electrical Engineering, Lakehead University, Barrie, ON, Canada E-mail: e-mail: wejaz@lakeheadu.ca

Clémentin Tayou D. [6]
Department of Computer science, IUT-Bandjoun, Cameroon.
E-mail: dtayou@gmail.com

Etienne Gnimpieba Zohim [7]
Department of Biomedical Engineering, University of South Dakota, USA.
E-mail: etienne.gnimpieba@usd.edu

1 Introduction

The fifth-generation (5G) wireless network marks the beginning of a truly digital society. It achieves significant breakthroughs in latency, data rates, mobility and number of connected devices, unlike previous generations [10, 41]. After completing the first comprehensive set of 5G standards, the initial commercial deployment of 5G wireless networks began in 2019. However, 5G will not meet all the future requirements in 2030+. Researchers are now focusing on sixth-generation (6G)

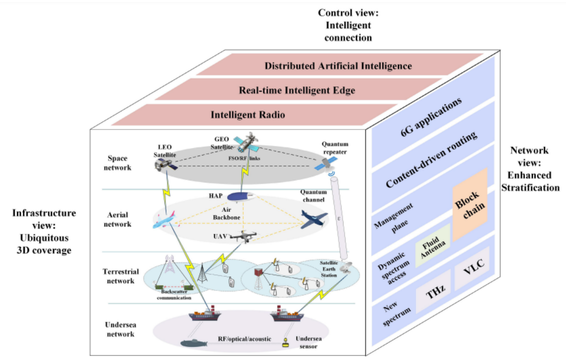


Fig. 1: Possible 6G communication architecture scenario. [17]

Fig. 2: Emerging 6G Applications. [9]

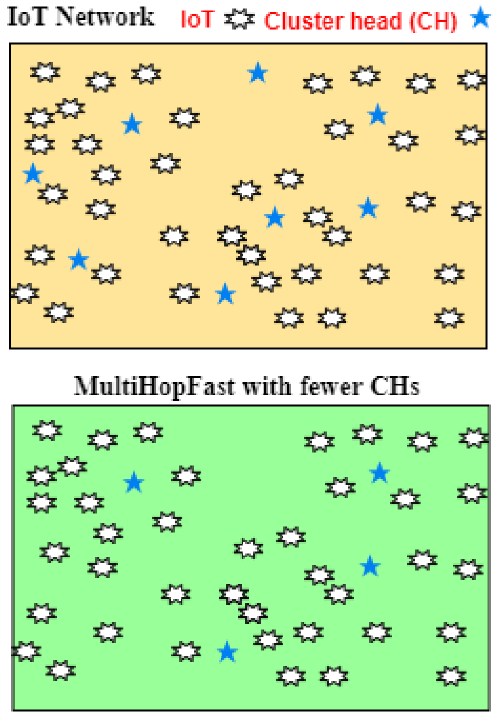


Fig. 3: IoT Network configuration with and without MultiHopFast

wireless communication networks. One of the key distinguishing features of 5G is low latency or more accurately guaranteed (deterministic) latency, which requires a deterministic network to ensure end-to-end latency with the timeliness and accuracy demanded by future disaster scenarios [19, 44, 51]. 6G will have additional high timing and phase synchronization accuracy requirements beyond what 5G can provide. Additionally, 6G will need to provide close to 100% geographic coverage, sub-centimetre geolocation accuracy, and millisecond geolocation update rate to meet the use case. 6G networks are expected to enable energy-efficient and socially transparent wireless connections globally. In contrast, the existing network architecture is capable of securing future delivery constraints of emerging applications that will leverage capabilities as illustrated in Fig. 2 and architectures on 1. It can also be predicted that many new applications emerge from the fusion of IoE (Internet of everything) and 6G communication [26]. However, IoE is expected to depend on 6G. It requires the ability to connect N smart devices where N term is scalable and can reach billions. Moreover, the Internet of Everything (IoE) also needs high data rates to support and facilitate the number of low latency devices [32]. Therefore, IoE and 6G can enable the business process to create data volumes and reinvent digitalization with improved and agile data analysis.

The sixth-generation (6G) wireless networks are envisioned to be robust, spectrally and energy-efficient, fast, reliable, and capable of supporting the massive connectivity of devices. These requirements are also aligned with exponential growth in the existing number of Internet-of-Things (IoT) devices [1, 7]. In recent years, a remarkable development in the network of connected objects such as smart devices, tablets, laptops, automobiles, and drones has been observed from the growth of the IoT [3, 53]. It continues to extend this unfailing progression in cities, corporate offices, campuses, healthcare, and even homes [3, 4, 12]. Devices can be quickly linked to networks and can collect data. IoTs are sensing devices that inherently consume less energy and possess limited computing power, and are capable of communicating with each other via the Internet [40]. Today's IoT network is an advance and more practical application of wireless sensor networks (WSN). In traditional WSNs, sensing nodes are connected to a router or any central node without a direct connection to the Internet. Whereas IoT nodes in modern IoT network applications are directly connected to the Internet for applications like portable devices, smart devices, dish machines, intelligent transportation systems [33,35]. These devices may be linked to a regular sink having the capacity to exchange information to each other [2,28,52]. Moreover, these entities are also able to communicate with human beings [45]. One of the essential roles of the IoT is listening and gathering information from the coverage area and transferring the gathered information to the sink node. This architecture is extensively replicated in many engineering sectors like military defence, emergency rescue, anti-terrorism, space exploration, national security, medical health, smart farming, smart cities, transportation, and surveillance. [13,18,42,47].

Authors in [50] defined a set of elements required for future IoT applications. Among other things security, is an essential requirement [16]. The IoT systems must support the interconnection of networks and communication between diverse applications and sustain interoperability. It is also important to highlight the quality of service, resource control, and energy optimization as outlined in the [50]. A problem related to managing resources is the balance of loads in the network. This is because the server may not manage all the individual connections when we are in a large-scale network. The microscale equipment is the bottleneck of the IoT network [50] because of limited resources such as processing and storage. For application services such as smart city or smart building management, IoT devices placed in a particular area can collect and transmit detected data or contextual data to servers on the Internet periodically or in real-time. The data size can vary from small such as tension or humidity, to relatively large, such as images. Grouping IoT objects can be a solution to maintain network stability insofar as choosing certain nodes for the transmission of data to the server.

We propose a clustering algorithm that preserves the structure of a large-scale network, such as in [23,43]. We assume that the communications between the two entities are reliable. The proposed approach tries to minimize the number of coordinators in the network. We obtain an average number of coordinators according to the number of nodes to balance the network structure by respecting the constraint of the number of hops between an IoT node and its coordinator. Improves the duration of the batteries of IoT terminals which can optionally be recharged or replaced in automatic IoT systems with mobile terminals. Finally, the algorithm runs in constant time, which has almost no impact on real-time IoT applications in health systems. As a result, computation-intensive routing protocols cannot be deployed on every IoT node. The inclusion of energy, load balancing, time complexity in clustering and routing protocols is still a great challenge. Clustering aims to partition the IoT network into groups to simplify operations like routing. Certain algorithms have already been proposed to solve the problem of clustering in WSNsr [8,29,30], vehicle networks [20,38], and IoT applications [43,49]. Fig. 3 shows a high-level view of IoT network configuration with and without proposed MultiHopFast algorithm.

1.1 Contributions

We proposed a clustering algorithm that preserves the structure of a large-scale network. We assume that the communications between the two entities are reliable. The proposed approach tries to minimize the number of coordinators in the network. We obtain an average number of coordinators according to the number of nodes to balance the network structure by respecting the constraint of the number of hops between an IoT node and its coordinator. Improves the duration of the batteries of IoT terminals which can optionally be recharged or replaced in automatic IoT systems with mobile terminals. Finally, the algorithm runs in constant time, which has almost no impact on real-time IoT applications in health systems. Following is the summary of contributions:

- We proposed a multi-hop clustering algorithm (MultiHopFast) for emerging IoT applications.
- The MultiHopFast algorithm proves to operate with a constant-time complexity.
- The MultiHopFast algorithm operates with a minimum number of coordinators (or clusters) and wireless connections.
- Operation of the MultiHopFast algorithm minimizes overall network energy consumption.
- Extensive simulations are conducted for the MultiHopFast algorithm and compared with state-of-the-art.

Rest of the paper is organized as follows: Section 2 presents the state of art, while section 3 addresses the proposed protocol, afterwards section 4 concerns the simulation results and finally section 5 conclude this paper.

2 Related work

Clustering algorithms are very often used to designate a self-organizing architecture in a network in order to facilitate operations such as data collection, and routing [6,11,36,46]. To compare these different clustering algorithms, authors frequently use metrics such as energy (the lifetime of network), distribution, scalability, the scale of the network, time complexity [14,20,23,37–

39, 43, 48]. In what follows, we present in-depth some relevant works from state-of-the-art.

In [46], the authors bring forward a multi-hop routing algorithm to minimize network load and consumed energy to increase the data transfer efficiency in the network. The process of electing the cluster heads (CHs) is done thanks to the remaining energy criterion of the IoTs. The computation of the intra-cluster path is done by transmitting the entire routing table forward to the CH. The results of this method show that it disperses the network load. However, it can be noted that this process is repeated each time a new node joins the network.. In addition, the election of a CH is greedy in energy consumption, and the conservation of routing tables is prohibitive for the IoT. Authors in [6] proposed a protocol named IoT-LEACH working with a random selection of CH at each turn. For all rounds, the clusters are formed using messages provided by CHs, and a CH changes at each round. If a CH has not used all of its energy during its turn, it may still have enough energy to continue as a CH in the next round. According to the LEACH protocol, it can no longer be CH. IoT LEACH solves this limit by defining a threshold in the existing LEACH protocol so that if a CH ends its turn with still more energy than the defined threshold, then it can continue to the next turn. This makes it possible to control the network's energy and thus minimize it. However, it only allows single-hop communications.

The authors in [39] proposed two methods of forming clusters. The first technique assumes that IoT devices have information about the lower layer. This method is used to conserve the energy of the detection layer over communication overhead. The network consists of a dominant set and the NP-complete problem of the maximum spanning tree. The node chosen as CH is the one with the maximum residual energy. The second one uses parameters like residual energy and the size of the neighbourhood. The node chosen as the CH has the most significant neighbourhood size and the highest residual energy. The cluster is therefore made up of the chief and its neighbours. These techniques conserve the network's energy, but the CH selection algorithm requires a vote and a graph traversal, respectively, for the first and second algorithms, representing a polynomial complexity and posing a problem for real-time applications in large-scale networks.

A clustering protocol for continuously powered IoT networks is proposed in [43]. The described approach aims to choose the minimum number of group leaders maximizing the number of jumps between ordinary nodes and their respective leaders. This is done in two stages. In the first step, Dijkstra's shortest path first algorithm is used to establish on the basis of the maximum hopping constraint H. The nodes having the possibility of communicating in H maximum hops from a source node. A CH is elected at the end of this phase. The second step is to select the best path to route information to the CH with the number of hops as a metric. This method makes it possible to reduce the connection load of the different servers; however, takes too long to run and would not be efficient in a real-time system.

Recently, the emergence of technologies such as LoRa has aroused the interest of many researchers around the world. The authors in [14] proposed a clustering algorithm based on LoRa technology. This algorithm divides the network into layers and in each layer, an intelligent gateway selection process is applied. This algorithm is compared to other protocols in the literature and proves that it is very energy efficient and with an interesting packet delivery rate. However, it still presents the problem of the formation time of multi-hop layers, which can be improved.

In summary, given in Table 1, some elements are still lacking in the existing protocols. For example, the authors [34, 37, 46] proposed protocols without taking into account the increasing evolution of networks. This prevents the network from growing to a certain extent. The mechanisms proposed in [6, 11, 38] are challenging to implement in large-scale networks. This limits these algorithms to small networks. In [5, 22, 23, 48], the authors presented mechanisms that have a high execution time. This can degrade the performance of applications in networks. Finally, the [39, 43] protocols do not sufficiently take into account the energy consumption of the proposed protocols, which is a problem when IoT nodes operate on battery power. In comparison, the proposed MultiHopFast is a mechanism that considers energy, scalability, time complexity and large scale for the IoT network.

3 A MultiHop constant time complexity clustering algorithm for IoT

This section presents a power-efficient and scalable load balancing protocol that efficiently runs in a large-scale network with constant time complexity. We introduce network modelling and analysis, expose our proposed mechanism, and assess its complexity.

3.1 Network modeling and analysis

We assume the network composed of a set designated by a linear representation $G = (V, E)$, where V is the set of elements in the graph $V = \{v_1, v_2, ..., v_n\}$ and E is the set of different direct connection between IoT devices

Table 1: Comparison of existing clustering algorithm.

Reference	Distributed	Scalability	Large Scale Network	Time Complexity	Energy	CH Determination	Objective
[46]	Yes	No	No	$O(1)$	Yes	Voting	Load balance
[6]	Yes	Yes	No	$O(1)$	Yes	pre-assign	Load balance
[11]	No	Yes	No	$O(1)$	Yes	Voting	Network lifetime
[38]	No	None	No	None	Yes	Voting	Mobility
[37]	Yes	No	No	$O(1)$	Yes	pre-assign	Load balance
[23]	Yes	Yes	No	$O(n)$	Yes	Voting	Network lifetime
[39]	No	Yes	No	$O(n)$	No	Voting	Network lifetime
[14]	Yes	Yes	Yes	$O(h)$	Yes	None	None
[43]	No	Yes	Yes	$O(n^2)$	No	Random	Load balance
[22]	Yes	Yes	No	$O(n)$	Yes	Voting	Network lifetime
[27]	No	Yes	No	None	Yes	None	Network lifetime
[34]	Yes	No	No	$O(n)$	Yes	Voting	Network lifetime
[5]	No	No	No	$O(n)$	Yes	Voting	Security
[48]	No	No	Yes	$O(n)$	Yes	Voting	Network lifetime
[21]	Yes	Yes	Yes	$O(1)$	Yes	Pre-assigne	Network lifetime

such as $\forall v_i, v_j \in V, \{v_i, v_j\} \subset V$, $dist(v_i, v_j) < TR_{v_i}$ or TR_{v_i} represents the transmission radius of the v_i node. Each node is able to play the role of coordinator. In MultiHopFast, each node determines in parallel its objective function f which depends on certain variables like the deployment surface, the number of nodes and the diffusion radius. This mechanism aims to find the set $C \subset V$ of CHs such that all the nodes of the network can reach the internet server via an element of $c_i \in C$ while respecting the constraint of the maximum hop number H. C is constructed in a distributed manner by determining beforehand a set $P \subset V$ such that

$$P = \{v_i, i = \{1, 2, ..., l\} | f_{v_i} > threshold\} \tag{1}$$

3.2 Proposed Mechanism

MultiHopFast consists of two stages.

3.3 First step

The opening stage builds the smallest set C of CHs in parallel, covering the whole network while constraining the number of hops h. The second stage is called reorganizing the network using the delay parameter on nodes from the first stage without the cluster. The smallest set of CHs C is determined by pre-assigning to each node a value calculated by its objective function using parameters like the number of possible CHs m, the network node number n and a constant β. m is determined according to the deployment surface a, the transmission radius of each node R_{v_i} and n. The function f is determined through the equation 2:

$$f = w_1 \times p + w_2 \times e, \tag{2}$$

where e and p represent remaining energy of IoT node and the likelihood of a node becoming CH respectively. Moreover, $w_1 + w_2 = 1$ and $(w_i, i = 1, 2)$ represents the weight attributed to each metric of the function f. The p is define by the equation 3 as follows [21]:

$$p = \beta \frac{m}{n}, \tag{3}$$

where β is a constant representing the redundancy coefficient which for a better efficiency takes the value (3) and m is determined by using [21]:

$$m = q \frac{a^2}{(r^2)}, \tag{4}$$

where

1. r is the transmission radius of each node.
2. a the deployment surface.
3. q a constant between $q_1 = \frac{1}{4\log(n)}$ and $q_2 = \frac{1}{\log(n)}$. A simplified demonstration can be seen in [21], where network surface area is divided into squares for the efficient operation of the network.

Let T be a constant value. Each object calculates its objective function. When $f \geq threshold$, the Algorithm 1 is executed otherwise the algorithm 2.

In the Algorithm 1, each v_i randomly chooses a number k between 1 and T. After k seconds, if the node receives a message, it checks the value of hop and exits from the algorithm when hop $= h$ or increments it and diffuses before exiting the algorithm. In this second option then there is a CH node. The second option allows each node to hold its relay to reach the CH.

In the Algorithm 2 executed by the $V \backslash \{P\}$. Each node waits for possible messages during clustering. When a message is received, each node checks the hop value x and increments x when it is less than H and saves its relay to the CH before exiting the algorithm. Otherwise, it saves the relay to the CH and leaves the algorithm.

3.4 Second step : Reorganization

After stage 1, one or more nodes may be neither members nor CHs. In this case, it is expected that these nodes execute the Algorithm 3. Each node generates a random number T1 and waits for the T1 number of

Algorithm 1 For nodes where $f_{v_j} \geq$ threshold.

Input Data : r, f_{v_i}.
Output Data : NF : Node function
For all $v_j \in V | f_{v_j} \geq threshold$
Choose number k between 1 and T.
Listen the network between 1 and k^{th}
if no MIO before k **then**
    Initialize a message
    h=1
    NF= CH,
    Broadcast message.
end if
if MIO message from $v_i (i \in N)$ **then**
    NF = Member
    $r = v_i$
    h++
    Diffuse message.
end if

Algorithm 2 For all other nodes.

Input Data : r, f_{v_i}.
Output Data : NF : Node Fonction.
Listen network
if message from v_j **then**
    NF= member
    h++
    Initialize new message and broadcast
end if

seconds. After that delay, if a node receives a message, it becomes a gateway and saves the relay to the CH. If a node does not receive a message, it becomes CH and informs the neighbourhood.

Algorithm 3 For all nodes who NF was not define after steps 1 or 2.

Input Data : r, f_{v_i}
Output Data : NF : Node function.
$\forall v_i \in V$
Choose random number k between 1 and T1
Listen network between 1^{th} and k^{th}
if MIO message **then**
    NF=CH
    h++
end if
if no message **then**
    Initialize message with h=1
    NF=CH
    Broadcast message
end if

3.5 example of simulation procedure

This subsection by figure 4 present different steps (4a,4b and 4c) simulation protocol.

3.6 Complexity analysis

Lemma 1 *Let E and V be the set of links and nodes of the IoT network. The entire process of building the smallest set C of CHs covering the whole network has a time complexity of $O(h)$. With h being the number of hops.*

Proof The proof of Lemma 1
Let h be the number of hops. In the Algorithm 1 of step 1, each node i with a $f_{v_i} \geq threshold$ must make a maximum of one broadcast each. The first level layer has just formed, and we have $h = 1$. The other network nodes' reception of a broadcast made by the layer checks the number of hops desired in the algorithm. If we have reached this number, we stop making broadcasts and exit with $h = 1$. Otherwise, we increment hop to signify that we have just formed layer 2 and $h = 2$. We proceed identically until we have at most h parallel broadcasts corresponding to k layers. So the first step gives us a complexity in $O(h)$.

Lemma 2 *Let E and V be the set of links and nodes of the IoT network. The reorganization process of the cluster has a time complexity of $O(h)$, With h being the number of hops.*

Proof The proof of Lemma 2
In the Algorithm 3 for the second step, the process is almost identical to that of Step 1 (Algorithms 1 and 2). The only difference is that in the Algorithm 3, all the nodes execute the same procedure. So using the proof of Lemma 1, we can deduce that the time complexity for the second step is $O(h)$.

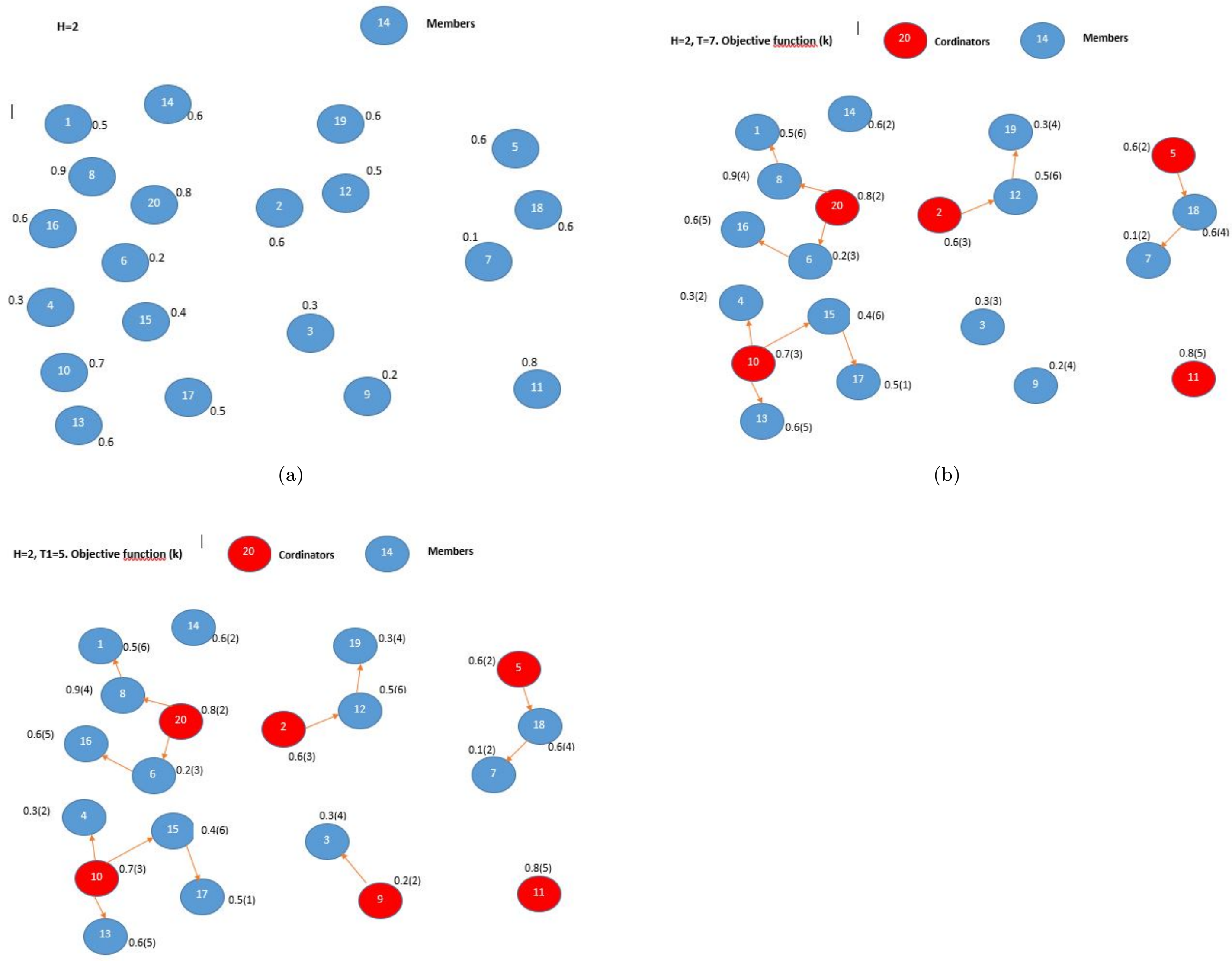
Theorem 1 *The temporal complexity of the described clustering protocol is in $O(h)$, with h being the number of hops.*

Proof The proof of theorem 1
The proof is trivial because it follows from the results of the proofs of Lemmas 1 and 2).

4 Simulation Results

The simulation is conducted on an i5 series processor with 4 GB of RAM using the Cooja network simulator and Contiki 3.0 OS. This simulator is a Java-based application that is capable of simulating large and small network sensor modules [31]. The simulation parameters are listed in Table 2.

The energy model we use is similar to the one in [15] that is E = ET+ER, where ET and ER are the energy used for the transmissions and the receptions of items in the network, respectively.



(a) (b) (c)

Fig. 4: Example of simulation using 20 nodes Iot

Table 2: Parameters for simulations settings.

Parameters	Min Temp
Network size	100m x 100m,25m*25m
Transmission radius	6,20,40,60,80 m
Number of nodes	5 to 120
Max. hop count constraint H	1,2,3,4,5 hops
Distribution of nodes	Random position
Radio medium	UDGM
Mote type	Sky mote
OS	Contiki 3.0

4.1 Results and discussions

Fig. 5 shows the evolution of time versus the number of IoT nodes for the clustering protocol [43] and the proposed approach. We have chosen [43] for the evaluation of our approach because it is state-of-the-art and close to the computationally optimal solution. This shows that our clustering is done by following a quasi-linear function $O(1)$ instead of in $\theta(n^2)$.

Fig. 6 presents the mean of selected coordinators by changing the amount of IoT nodes with a random deployment on a surface of 25m $\times$ 25m, a transmission radius TR $=$ 6m and nodes varying between 20-80 nodes. It also shows that the amount of selected coordinators is converging towards a given value rather than continuously rising. In other words, in the described protocol, it appears that relatively a backbone of fewer coordinators can furnish the Internet connection to the whole IoT devices in a specific period. As device den-

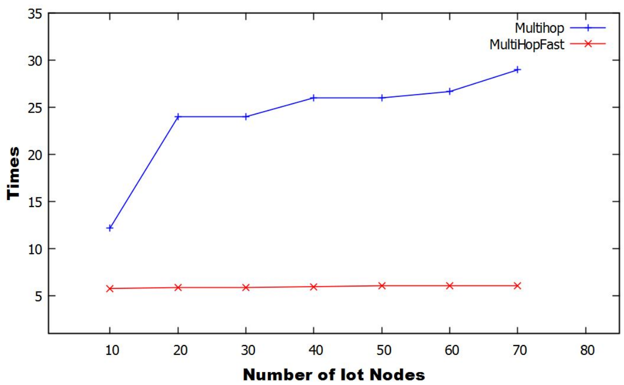


Fig. 5: Comparison of temporal performance according to number of nodes.

sity grows, the connectivity's degree between the devices becomes bigger and the size of the network, as shown in Fig. 6b. Moreover, in the latter figure, we can observe that the ratio of selected coordinators among all the network devices becomes lower as device density increases.

With a number of nodes from 5 to 80, an area of 25 × 25m and a radius of 6m, Figs. 7(a)-(e) indicate the average consumed energy of the whole network according to a number of hops from 1 to 5 respectively. Generally, IoT applications are limited to a small number of hops compared to traditional computer networking applications. For example, IoT devices in a house could consist of a few hops only. Therefore, we chose up to 5 hops. We observe that the average energy consumed by the network for the clustering process is less than the one in [43]. This shows that MultiHopfast is an energy-efficient mechanism compared to the latter [43]. This result also reveals that the number of hops in the network does not significantly influence the network's energy consumption in general. This shows that the number of hops is independent of the network's average energy consumption.

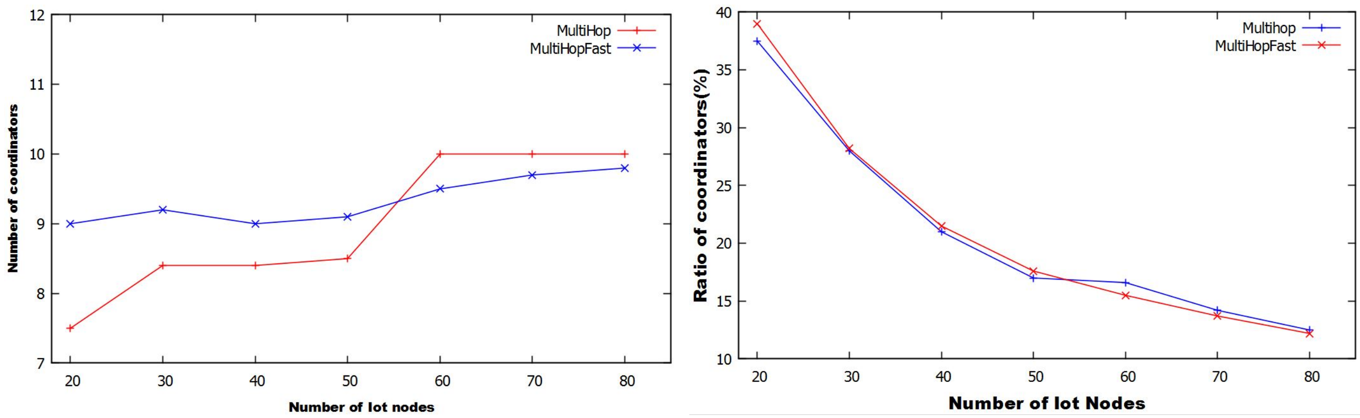
Fig. 8 presents the threshold's impact on the number of clusters and the number of nodes per cluster. We have varied the number of IoT nodes from 20 to 80 represented in Figs. 8a to 8d with a random deployment on a surface of 25m × 25m, a transmission radius TR = 6m. This result shows that when the chosen threshold is close to half, after an average of five simulations, we observe an average number of clusters and a balanced distribution of the member nodes for each cluster formed. This shows a balanced charge in the network. This study allows the following conclusions to be drawn: the threshold values chosen depend on the network structure chosen. When the threshold is high, there are fewer clusters, which corresponds to networks with very high transmission powers. However, when the threshold is very low, we have many clusters, which can be used when the network communications are done step by step.

The eq. (2) is the weighted sum of remaining energy e of an IoT node, and p is the likelihood of that node to become a CH with their associated weights w_2 and w_1 respectively. Note that $w_1 + w_2 = 1$. Fig. 9 shows the impact of the weights w_2 and w_1 of the objective function on the number of clusters formed. We notice that the number of clusters is almost constant according to the number of nodes with variations in weights w_2 and w_1. In other words, the proposed clustering algorithm show that an increase in the number of IoT nodes kept the number of cluster stable.

Figs. 10 (a)-(d) was obtained using the same network settings as for the Fig. 8 for the first stride of the clustering procedure. We note that the threshold defined an influence on the value of the percentage of participation of the nodes for this stage. When the threshold is very small, we observe that we have a large percentage of nodes participating in it. With a small value of the number of hops h the cluster formation is almost done, and the number of clusters is high. When the threshold is low, the participation rate in the first stride of the clustering procedure is low, with a very low number of clusters. The participation rate for the first stage of MultiHopFast is high with a small value of h and very low with a high value h.

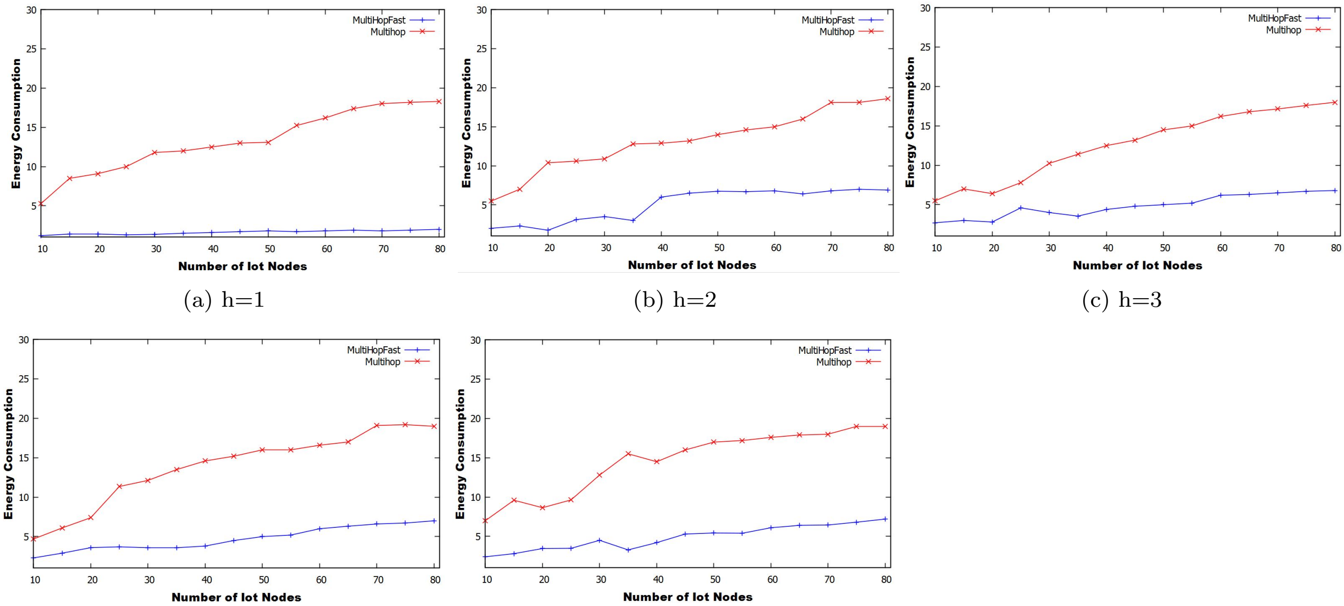
Figs. 11(a)-(d) present the evolution of the number of clusters according to the number of nodes in the network from 20 to 120 while varying the transmission radius of the sensors. We note that even when the transmission radius of the nodes is very large we always observe an average number of clusters, whatever the number of sensors varying on hops from 2 to 5. This shows that MultiHopFast is a protocol on which the variation the number of sensors. Therefore, it highlights its evolutionary character. Likewise, the transmission radius varying from 20 to 80 has no impact on the network's structure. This highlights the fact that MultiHopFast is suitable for large-scale networks.

Figs. 12 (a)-(d) present the evolution of the ratio of clusters according to the number of IoT nodes while respecting the constraint of the number of hops h for the transmission radio TR = 20, 40, 60, and 80. We note that the only case in which the number of clusters is high is when the number of hops is 1. In all other cases, the ratio of clusters is average, which is good in terms of balance. When there are too many CHs, we easily find clusters with only one or two nodes, which is almost unnecessary energy expenditure. When this number is too small, it is easy to see a CH that represents a very



(a) (b)

Fig. 6: Performances comparison between MultiHopFast and MultiHop [43] on (a) number of coordinators and (b) selected coordinator ratio.



(a) h=1 (b) h=2 (c) h=3

(d) h=4 (e) h=5

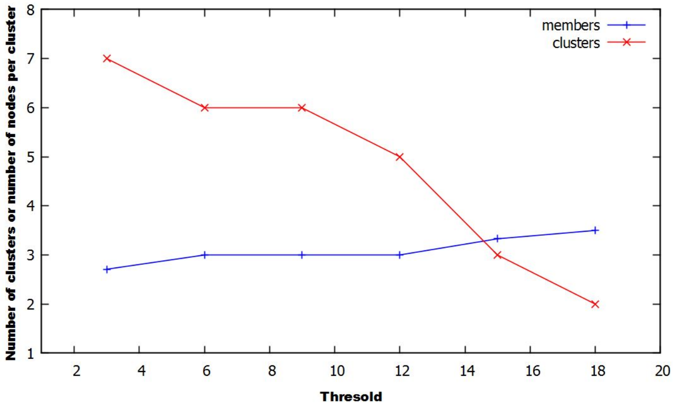
Fig. 7: Comparison of the average energy consumption between MultiHopFast and MultiHop (a) as a function of the number of IoT nodes for hop counts h from (a) 1, (b) 2, (c) 3, (d) 4, and (e) 5.

large number of nodes at a time. This result shows that MultiHopFast achieved its goal of load balancing for nodes in the network.

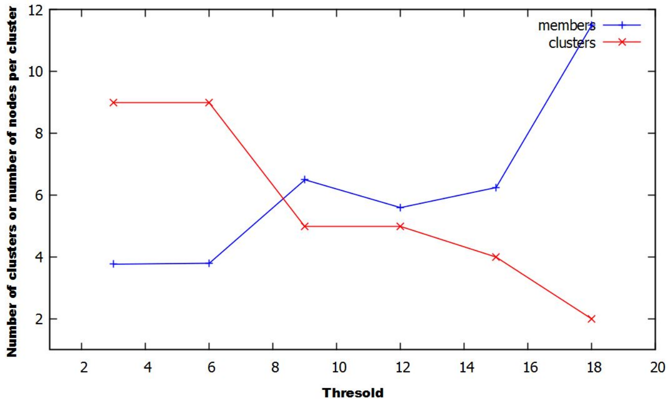
Figs. 13(a)-(d) number of member nodes per CH versus the number of IoT nodes for transmission radio ranging from 20 to 80, respectively. We notice that the number of member nodes does not evolve exponentially. When the hopping number $h = 1$, the number of member nodes is very small, which implies that the number of coordinators is very large. For $h = 2$ the number of nodes is more significant, which means that the number of coordinators is small. These results confirm the results obtained in Fig. 11, the number of clusters formed is neither too high nor too low to save as much energy as possible.

5 Conclusion and future work

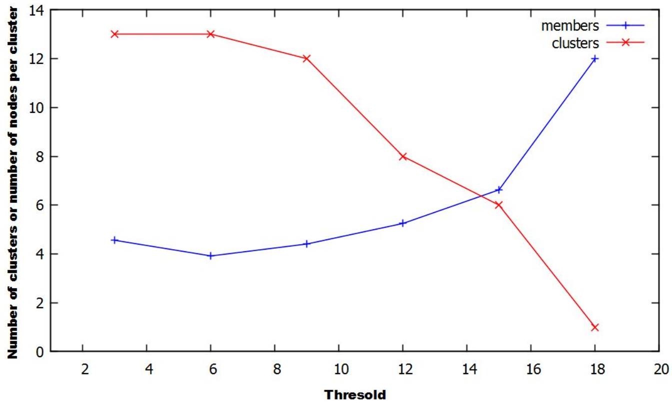
This paper presented a constant-time complexity clustering MultiHopFast algorithm for emerging IoT applications. The overall goal of the MultiHopFast algorithm is to address the inherent IoT network challenges like meagre computing resources, load balancing, and



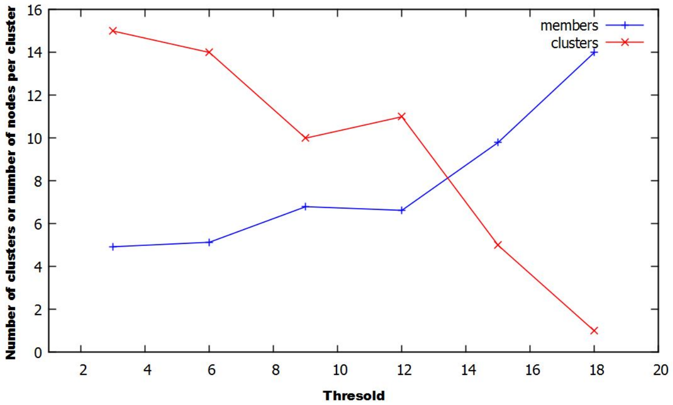
(a) Number of Iot nodes (20)



(b) Number of Iot nodes (40)

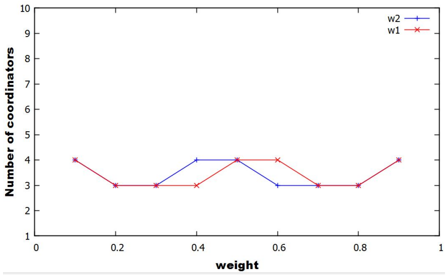


(c) Number of Iot nodes (60)

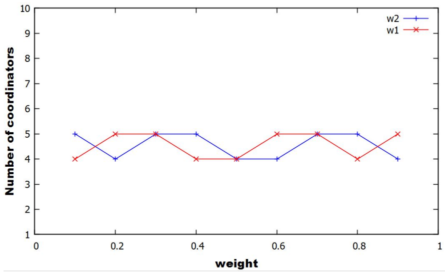


(d) Number of Iot nodes (80)

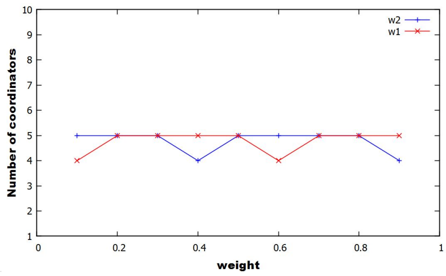
Fig. 8: Impact of threshold on number of clusters and number of IoT nodes per cluster.



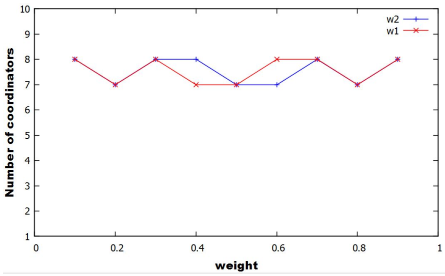
(a) Number of Iot nodes (20)



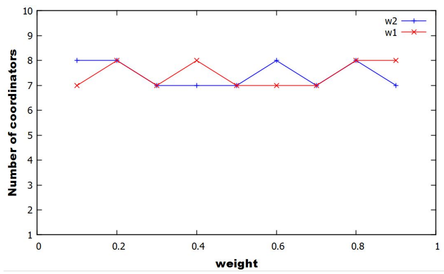
(b) Number of Iot nodes (40)



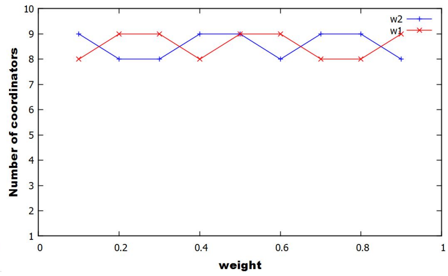
(c) Number of Iot nodes (60)



(d) Number of Iot nodes (80)

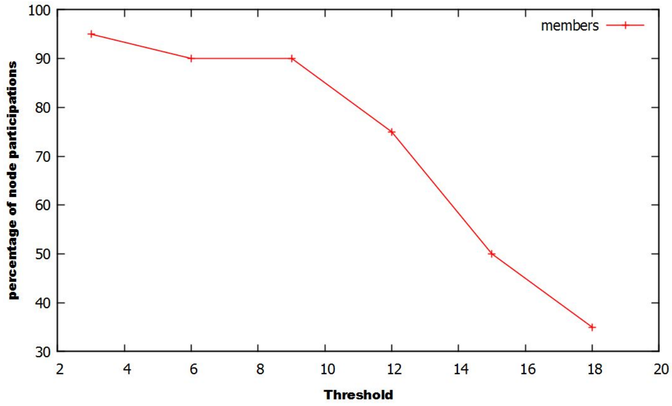


(e) Number of Iot nodes (100)

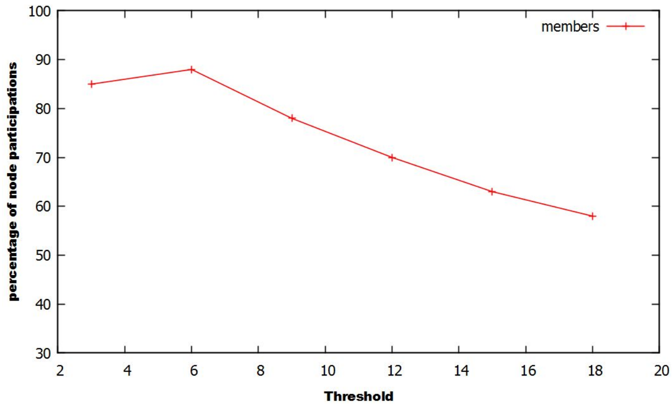


(f) Number of Iot nodes (120)

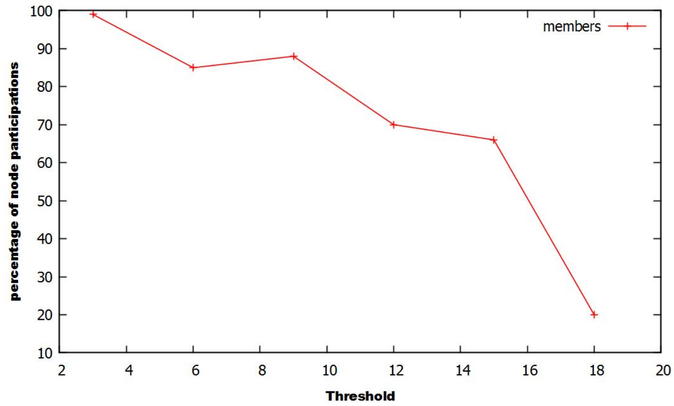
Fig. 9: Impact of the weight value (w_1 and w_2) on the number of clusters for IoT nodes from 20 (9a) to 120 (9f).



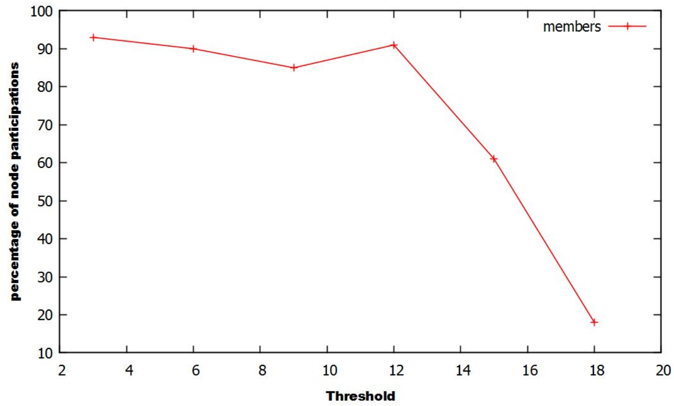
(a) Number of Iot nodes (20)



(b) Number of Iot nodes (40)



(c) Number of Iot nodes (60)



(d) Number of Iot nodes (80)

Fig. 10: Impact of the threshold on the participation of nodes in the first step of MultiHopFast

network scalability with time efficiency. The MultiHopFast algorithm is a multi-hop clustering scheme dedicated to IoT network applications to reduce internet connections' active (or operative) numbers. This algorithm minimized the number of coordinators (or cluster heads (CHs)) representing associated clusters. The MultiHopFast algorithm also used a minimal number of hops between a CH and cluster members (i.e., IoTs). As a result, this algorithm operated with constant-time complexity and created up to 12% fewer CHs to save energy as a critical resource. Our simulation results used well-known performance criteria, which showed that the MultiHopFast algorithm was optimal when compared against the existing state-of-the-art algorithms. However, MultiHopFast lacks a mechanism for managing IoT applications in mobile environments.

In the future, we would like to implement this constant-time complexity algorithm for mobile IoT applications and unreliable communication scenarios. We further tend to improve our clustering algorithm using some existing clustering techniques like [24] and [25].

Conflict of Interest The authors declare no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Informed consent was obtained from all individual participants included in the study.

Author contributions This work was conceptualized and designed by Yann and Alain. Experiment was designed by Yann, Alain, Hafiz and David. Initial draft was prepared by Yann and Alain. After initial draft, Hafiz, Yann, Alain, David, Clémentin, Etienne, Waleed give input to improve quality and presentation. Etienne is PI of this work and edited first and subsequent draft of the manuscript. All the authors read and approved the final manuscript.

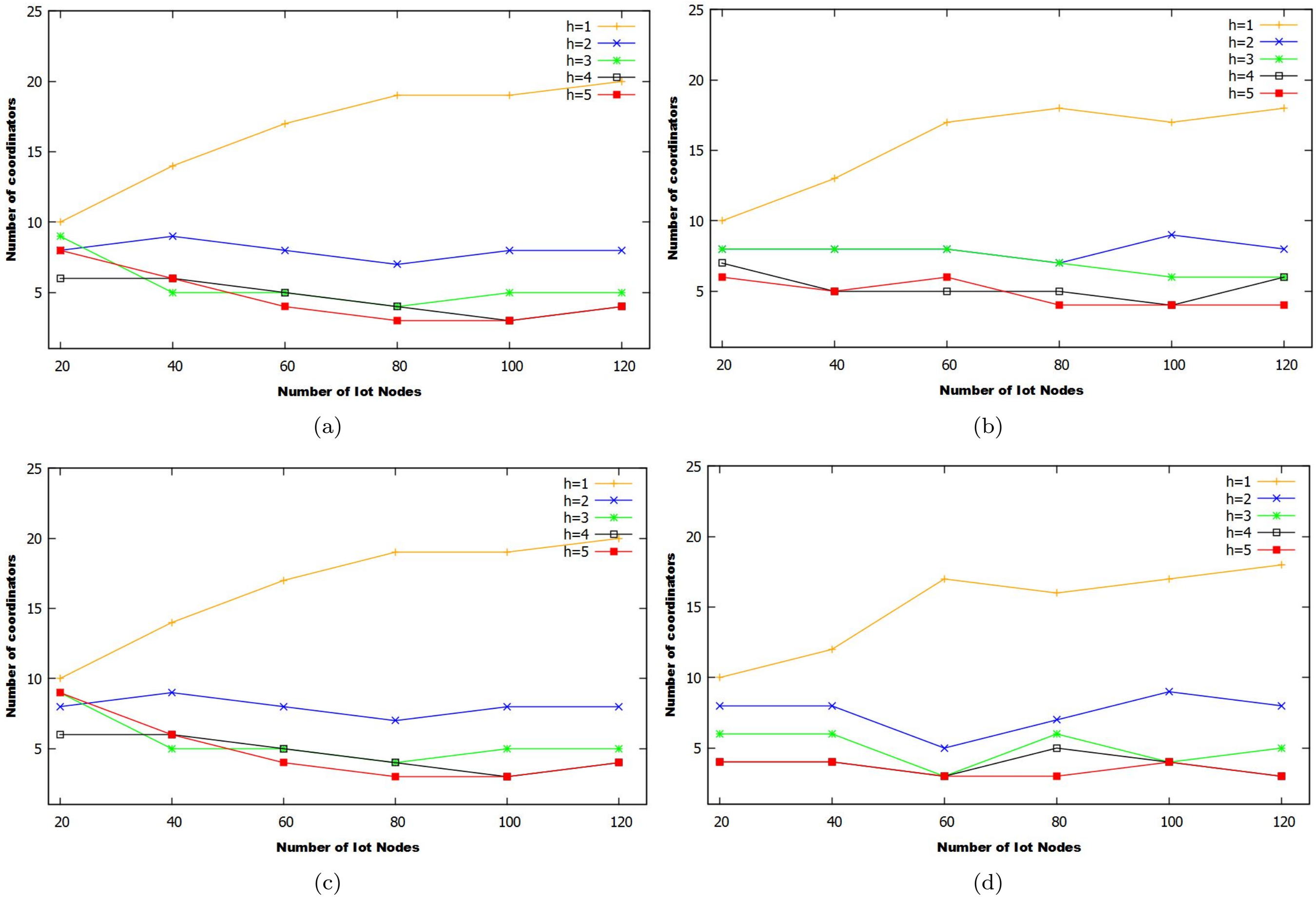


Fig. 11: Performance comparison for different h values=$\{1,2,3,4,5\}$ according to the number of coordinates under different $|N|$; for 11a-11d TR=20, 40, 60, 80 respectively.

Data availability The present study is based on synthesized data generated randomly by the authors based on some parameters mentioned in the above text.

References

1. M. W. Akhtar, S. A. Hassan, R. Ghaffar, H. Jung, S. Garg, and M. S. Hossain. The shift to 6g communications: vision and requirements. *Human-centric Computing and Information Sciences*, 10(1):1–27, 2020.
2. G. A. Akpakwu, B. J. Silva, G. P. Hancke, and A. M. Abu-Mahfouz. A survey on 5g networks for the internet of things: Communication technologies and challenges. *IEEE Access*, 6:3619–3647, 2017.
3. H. M. Ali, J. Liu, S. A. C. Bukhari, and H. T. Rauf. Planning a secure and reliable iot-enabled fog-assisted computing infrastructure for healthcare. *Cluster Computing*, pages 1–19, 2021.
4. H. M. Ali, J. Liu, and W. Ejaz. Planning capacity for 5g and beyond wireless networks by discrete fireworks algorithm with ensemble of local search methods. *EURASIP Journal on Wireless Communications and Networking*, 2020(1):1–24, 2020.
5. D. Amine, B. Nasr-Eddine, and L. Abdelhamid. A distributed and safe weighted clustering algorithm for mobile wireless sensor networks. *Procedia Computer Science*, 52:641–646, 2015.
6. T. M. Behera, U. C. Samal, and S. K. Mohapatra. Energy-efficient modified leach protocol for iot application. *IET Wireless Sensor Systems*, 8(5):223–228, 2018.
7. A. B. Bomgni, G. B. J. Mdemaya, H. M. Ali, D. G. Zanfack, and E. G. Zohim. Espina: efficient and secured protocol for emerging iot network applications. *Cluster Computing*, 2022.
8. A. B. BOMGNI and Y. B. C. MTOPI. Calibrated clustering algorithm for mobile ad hoc network. *International Journal of Computer Science and Information Security (IJCSIS)*, 17(3), 2019.
9. C. De Alwis, A. Kalla, Q.-V. Pham, P. Kumar, K. Dev, W.-J. Hwang, and M. Liyanage. Survey on 6g frontiers: Trends, applications, requirements, technologies and future research. *IEEE Open Journal of the Communications Society*, 2:836–886, 2021.
10. C. De Lima, D. Belot, R. Berkvens, A. Bourdoux, D. Dardari, M. Guillaud, M. Isomursu, E.-S. Lohan, Y. Miao, A. N. Barreto, et al. Convergent communication, sensing and localization in 6g systems: An overview of technologies, opportunities and challenges. *IEEE Access*, 9:26902–26925, 2021.

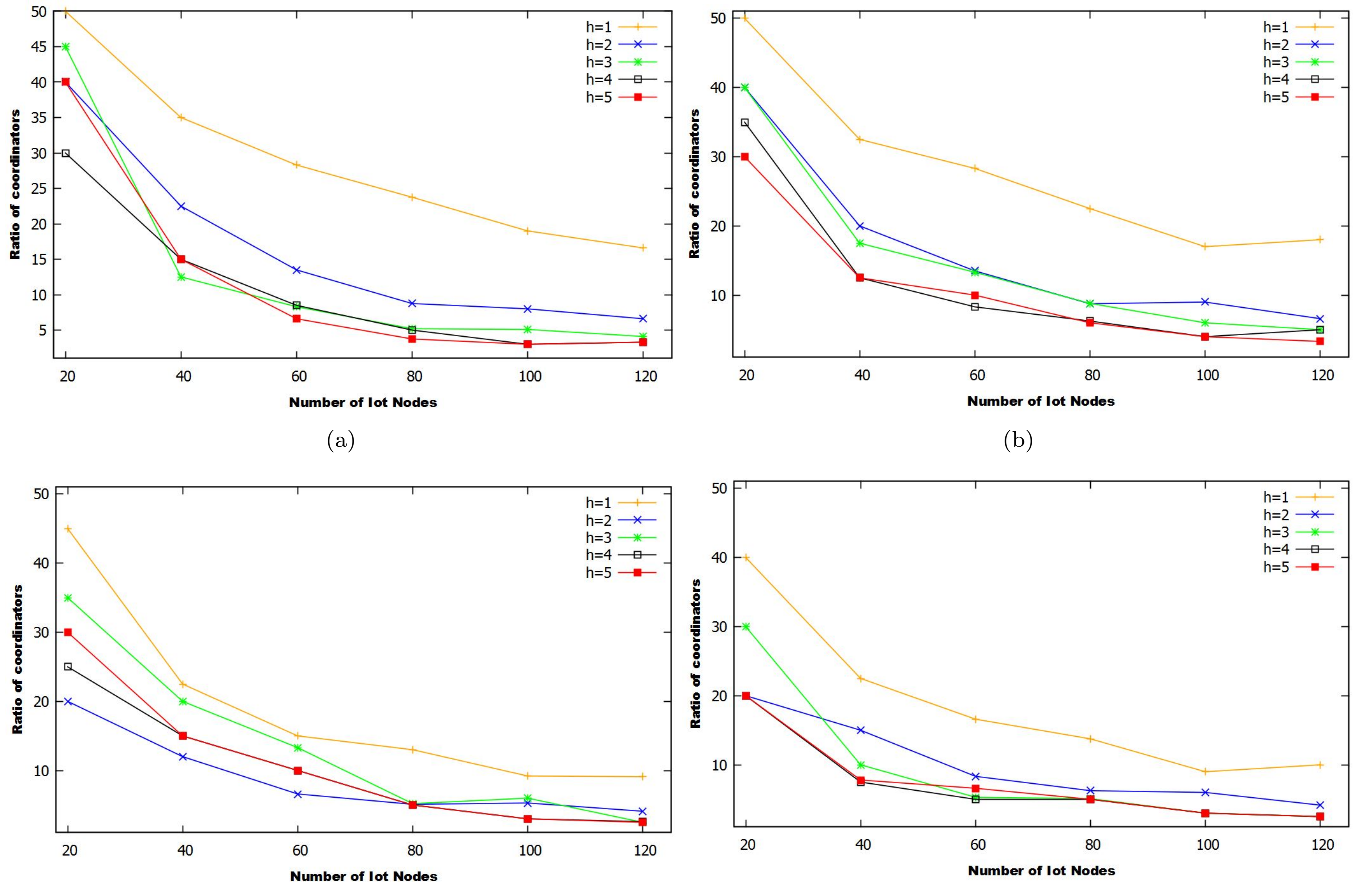


Fig. 12: Compared performances for different h values $=\{1,2,3,4,5\}$ according to the ratio of coordinators under different $|N|$ for (12a-12d) TR=20, 40, 60, 80 respectively.

11. M. Elhoseny, A. Farouk, N. Zhou, M.-M. Wang, S. Abdalla, and J. Batle. Dynamic multi-hop clustering in a wireless sensor network: Performance improvement. *Wireless Personal Communications*, 95(4):3733–3753, 2017.
12. H. M. A. et al. Optimising the power using firework-based evolutionary algorithms for emerging iot applications. *IET Networks*, 8(1):15–31, 2019.
13. A.-T. Fadi and A. Sinem. Context-sensitive access in industrial internet of things (iiot) healthcare applications. *IEEE Transactions on Industrial Informatics*, 14(6):2736–2744, 2018.
14. M. O. Farooq. Clustering-based layering approach for uplink multi-hop communication in lora networks. *IEEE Networking Letters*, 2(3):132–135, 2020.
15. W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan. Energy-efficient communication protocol for wireless microsensor networks. In *Proceedings of the 33rd annual Hawaii international conference on system sciences*, pages 10–pp. IEEE, 2000.
16. J. L. Hernández-Ramos, J. B. Bernabé, and A. Skarmeta. Army: architecture for a secure and privacy-aware lifecycle of smart objects in the internet of my things. *IEEE Communications Magazine*, 54(9):28–35, 2016.
17. T. Huang, W. Yang, J. Wu, J. Ma, X. Zhang, and D. Zhang. A survey on green 6g network: Architecture and technologies. *IEEE access*, 7:175758–175768, 2019.
18. J. Jagannath, N. Polosky, A. Jagannath, F. Restuccia, and T. Melodia. Machine learning for wireless communications in the internet of things: A comprehensive survey. *Ad Hoc Networks*, 93:101913, 2019.
19. W. Jiang, B. Han, M. A. Habibi, and H. D. Schotten. The road towards 6g: A comprehensive survey. *IEEE Open Journal of the Communications Society*, 2:334–366, 2021.
20. M. F. Khan, F. Aadil, M. Maqsood, S. H. R. Bukhari, M. Hussain, and Y. Nam. Moth flame clustering algorithm for internet of vehicle (mfca-iov). *IEEE Access*, 7:11613–11629, 2018.
21. L. Kong, Q. Xiang, X. Liu, X.-Y. Liu, X. Gao, G. Chen, and M.-Y. Wu. Icp: Instantaneous clustering protocol for wireless sensor networks. *computer Networks*, 101:144–157, 2016.
22. H. Kour and A. K. Sharma. Hybrid energy efficient distributed protocol for heterogeneous wireless sensor network. *International Journal of Computer Applications*, 4(6):1–5, 2010.
23. J. S. Kumar and M. A. Zaveri. Clustering approaches for pragmatic two-layer iot architecture. *Wireless Communications and Mobile Computing*, 2018, 2018.
24. Z. Li, F. Nie, X. Chang, L. Nie, H. Zhang, and Y. Yang. Rank-constrained spectral clustering with flexible embed-

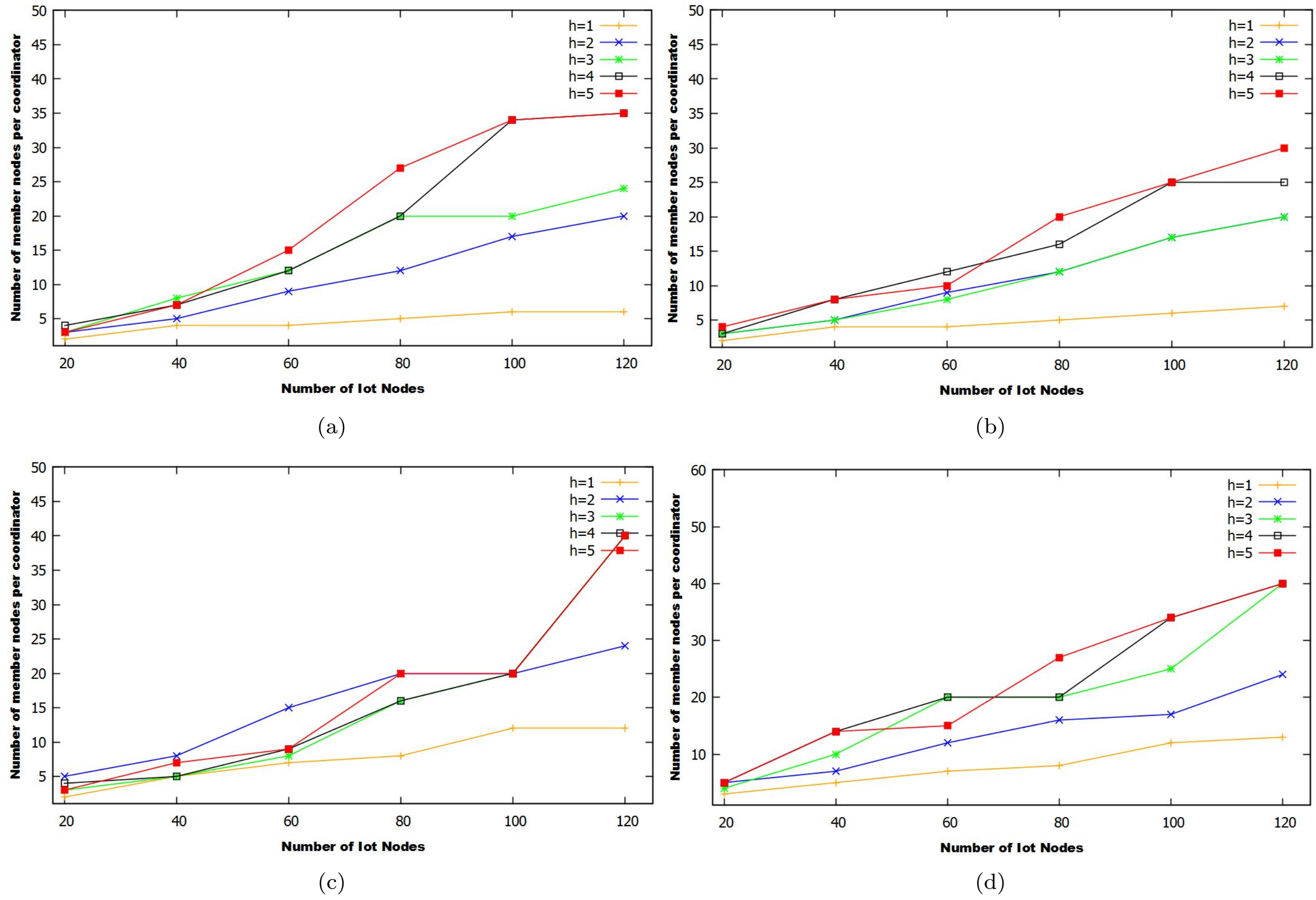


Fig. 13: Compared performances for different h values = 1,2,3,4,5 according to the average number of member nodes for a coordinator under different $|N|$; for 13a-13d TR={20, 40, 60, 80} respectively.

ding. *IEEE Transactions on Neural Networks and Learning Systems*, 29(12):6073–6082, 2018.

25. Z. Li, F. Nie, X. Chang, Y. Yang, C. Zhang, and N. Sebe. Dynamic affinity graph construction for spectral clustering using multiple features. *IEEE Transactions on Neural Networks and Learning Systems*, 29(12):6323–6332, 2018.
26. Z. Lv and N. Kumar. Software defined solutions for sensors in 6g/ioe. *Computer Communications*, 153:42–47, 2020.
27. A. Mahajan, K. Sharma, and M. Scholar. An approach towards unequal clustering in wireless sensor networks. *International Journal of Engineering Science*, 13195, 2017.
28. G. Manogaran, R. Varatharajan, D. Lopez, P. M. Kumar, R. Sundarasekar, and C. Thota. A new architecture of internet of things and big data ecosystem for secured smart healthcare monitoring and alerting system. *Future Generation Computer Systems*, 82:375–387, 2018.
29. M. Meddah, R. Haddad, and T. Ezzedine. An energy efficient and density control clustering algorithm for wireless sensor network. In *2017 13th International Wireless Communications and Mobile Computing Conference (IWCMC)*, pages 357–364. IEEE, 2017.
30. H. Mohapatra and A. K. Rath. Fault tolerance in wsn through pe-leach protocol. *IET wireless sensor systems*, 9(6):358–365, 2019.
31. K. P. Naik and U. R. Joshi. Performance analysis of constrained application protocol using cooja simulator in contiki os. In *International Conference on Intelligent Computing, Instrumentation and Control Technologies*, pages 547–550. IEEE, IEEE, 2017.
32. S. Nayak and R. Patgiri. 6g communication technology: A vision on intelligent healthcare. In *Health Informatics: A Computational Perspective in Healthcare*, pages 1–18. Springer, 2021.
33. M. R. Palattella, M. Dohler, A. Grieco, G. Rizzo, J. Torsner, T. Engel, and L. Ladid. Internet of things in the 5g era: Enablers, architecture, and business models. *IEEE Journal on Selected Areas in Communications*, 34(3):510–527, 2016.
34. M. Patil and R. C. Biradar. Energy efficient weighted clustering algorithm in wireless sensor networks. *Global Journal of Computer Science and Technology*, 17(2), 2017.
35. R. Pratim. A survey on internet of things architectures. *Journal of King Saud University-Computer and Information Sciences*, 30(3):291–319, 2018.
36. D. Reddy. A review-efficiency of energy clustering and routing in wireless sensor networks. *Int J Adv Technol*, 10(222):2, 2019.
37. R. A. Sadek. Hybrid energy aware clustered protocol for iot heterogeneous network. *Future Computing and Informatics Journal*, 3(2):166–177, 2018.
38. O. Senouci, Z. Aliouat, and S. Harous. Mca-v2i: A multi-hop clustering approach over vehicle-to-internet commu-

nication for improving vanets performances. *Future Generation Computer Systems*, 96:309–323, 2019.
39. S. K. Singh, P. Kumar, J. P. Singh, and M. A. A. Alryalat. An energy efficient routing using multi-hop intra clustering technique in wsns. In *TENCON 2017-2017 IEEE Region 10 Conference*, pages 381–386. IEEE, IEEE, 2017.
40. A. Sivanathan, H. H. Gharakheili, and V. Sivaraman. Inferring iot device types from network behavior using unsupervised clustering". In *IEEE LCN*. IEEE, 2019.
41. D. Soldani and A. Manzalini. Horizon 2020 and beyond: On the 5g operating system for a true digital society. *IEEE Vehicular Technology Magazine*, 10(1):32–42, 2015.
42. Z. Sun, X. Xing, T. Wang, Z. Lv, and B. Yan. An optimized clustering communication protocol based on intelligent computing in information-centric internet of things. *IEEE Access*, 7:28238–28249, 2019.
43. Y. Sung, S. Lee, and M. Lee. A multi-hop clustering mechanism for scalable iot networks. *Sensors*, 18(4):961, 2018.
44. H. Tataria, M. Shafi, A. F. Molisch, M. Dohler, H. Sjöland, and F. Tufvesson. 6g wireless systems: Vision, requirements, challenges, insights, and opportunities. *Proceedings of the IEEE*, 109(7):1166–1199, 2021.
45. S. Umamaheswari and A. Negi. Internet of things and rpl routing protocol: A study and evaluation. In *2017 International Conference on Computer Communication and Informatics (ICCCI)*, pages 1–7. IEEE, IEEE, 2017.
46. C. Wang, Y. Zhang, X. Wang, and Z. Zhang. Hybrid multihop partition-based clustering routing protocol for wsns. *IEEE Sensors Letters*, 2(1):1–4, 2018.
47. T. Wang, M. Z. A. Bhuiyan, G. Wang, M. A. Rahman, J. Wu, and J. Cao. Big data reduction for a smart city's critical infrastructural health monitoring. *IEEE Communications Magazine*, 56(3):128–133, 2018.
48. P. G. Wojciech Bednarczyk. An enhanced algorithm for manet clustering based on weighted parameters. *Universal Journal of Communications and Network*, 1(3):88–94, 2013.
49. J. Xiong, J. Ren, L. Chen, Z. Yao, M. Lin, D. Wu, and B. Niu. Enhancing privacy and availability for data clustering in intelligent electrical service of iot. *IEEE Internet of Things Journal*, 6(2):1530–1540, 2018.
50. I. Yaqoob, E. Ahmed, I. A. T. Hashem, A. I. A. Ahmed, A. Gani, M. Imran, and M. Guizani. Internet of things architecture: Recent advances, taxonomy, requirements, and open challenges. *IEEE wireless communications*, 24(3):10–16, 2017.
51. X. You, C.-X. Wang, J. Huang, X. Gao, Z. Zhang, M. Wang, Y. Huang, C. Zhang, Y. Jiang, J. Wang, et al. Towards 6g wireless communication networks: Vision, enabling technologies, and new paradigm shifts. *Science China Information Sciences*, 64(1):1–74, 2021.
52. Y. Zhang and Y. Wang. A novel energy-aware bio-inspired clustering scheme for iot communication. *Journal of Ambient Intelligence and Humanized Computing*, pages 1–10, 2020.
53. Y. Zhu, K. Chevalier, X. Wang, and N. Wang. Efficient mobile edge computing for mobile internet of thing in 5g networks. In *Proceedings of the 53rd Hawaii International Conference on System Sciences*, 01 2020.